

EBU6304 Software Engineering Group Project

2025-26

Group number		
member 1	QM no: 231222992	Name: Qingshang Yin
member 2	QM no: 2312229891	Name: Zixi Fang
member 3	QM no: 231222350	Name: Shuhan Yang
member 4	QM no: 231222110	Name: Jiayi Guo
member 5	QM no: 231220862	Name: Yuman Lv
member 6	QM no: 231220161	Name: Leyi Zhang

First report

1. Requirements

Describe the fact-finding techniques, iteration plan, prioritisation and estimation methods used in the project.

1. Requirements

1.1 Fact-finding techniques

At the beginning of the project, the team carried out requirements elicitation to understand the needs of the three main stakeholder groups in the system: Teaching Assistants (TAs), Module Organisers (MOs), and Admin staff. Since these three groups have different goals and responsibilities, we adopted more than one fact-finding technique in order to gather both broad and role-specific requirements.

The primary techniques used were **questionnaires** and **email enquiries**. Questionnaires were used to collect information from a wider set of potential users efficiently. This method was suitable for identifying common expectations, such as what information TAs want to include in their profile, what job details MOs need to publish, and what kinds of monitoring information Admin users require. The questionnaire responses helped the team identify repeated needs and prioritise the most common functions.

Email enquiries were used as a follow-up technique to clarify ambiguous answers and obtain more detailed feedback from selected stakeholders. Compared with questionnaires, email communication was more flexible and allowed the team to ask targeted questions about workflows, exceptions, and practical concerns. For example, email follow-ups were useful for confirming how MOs want to review applicants, how application status should be displayed to TAs, and what kind of workload overview would be meaningful for Admin users.

The team combined the findings from these two methods and translated them into a structured set of user stories. This process helped us move from raw stakeholder opinions to

implementable requirements. The collected evidence also supported the identification of three role-based functional groups: TA functions related to job seeking and self-presentation, MO functions related to job posting and candidate selection, and Admin functions related to workload monitoring and fairness control.

Overall, the use of both questionnaires and email enquiries improved the reliability of the elicitation process. Questionnaires provided breadth, while email enquiries provided depth and clarification. This combination ensured that the resulting backlog was not based on assumptions alone, but on direct stakeholder input.

1.2 Iteration plan

After gathering the initial requirements, the team organised the work into an iterative development plan. The backlog was divided into a series of sprints so that the system could be built progressively, starting from core end-to-end functionality and then expanding to management and enhancement features.

The iteration plan was designed around delivering a minimum viable workflow first. In this project, the essential workflow is: a TA creates a profile and uploads a CV, an MO publishes a job, the TA browses and applies for the job, and the MO reviews and updates the application outcome. Admin functions were scheduled slightly later because they depend on the existence of recruitment and hiring data generated by the earlier workflow.

Sprint 1 focuses on foundational features. These include account access, TA profile creation, CV upload, and MO job creation and publishing. The purpose of this sprint is to establish the minimum system structure required to support later interactions.

Sprint 2 focuses on the main recruitment interaction. In this sprint, TAs can browse, search, and apply for jobs, while MOs can edit and close job postings and begin to view submitted applicants. At the end of this sprint, the core recruitment flow should already be usable.

Sprint 3 focuses on selection and application progress. In this sprint, MOs can shortlist, accept, or reject applicants, and TAs can track application outcomes and receive notifications. This completes the decision-making side of the hiring process.

Sprint 4 focuses on administrative oversight. Once enough recruitment data exists, Admin users can view TA workloads, identify overloaded students, and monitor job distribution and application statistics across the system.

Sprint 5 focuses on lower-priority management and enhancement functions, such as user account administration, report generation, and audit logging. These features add governance and maintainability, but they are not required for the initial operational release.

This iteration plan was chosen because it reduces delivery risk. High-value and high-dependency functions are implemented first, while supporting and monitoring features are added later. It also allows feedback from early sprints to inform later refinements.

1.3 Prioritisation method

The team prioritised requirements using a **five-level priority scale: very high, high, medium, low, and very low**. This approach was selected because it is simple, easy to communicate, and suitable for an early-stage backlog where the main concern is deciding implementation order rather than performing highly formal ranking.

Several criteria were considered when assigning priorities.

First, we considered **business and user value**. Features that are necessary for the platform's core purpose received the highest priority. For example, TA profile creation, CV upload, job browsing, job application, MO job posting, and applicant decision functions were all treated as very high priority because without them the system cannot support recruitment.

Second, we considered **dependency relationships**. Some stories must be implemented before others can function properly. For instance, a TA cannot apply for a job before a profile exists and a job has been published. Similarly, Admin workload monitoring depends on application and hiring data being created first. These dependency relationships strongly influenced the sprint allocation.

Third, we considered **frequency of use**. Functions used regularly by TAs and MOs in the main workflow were placed above features that are only occasionally used by Admins or only useful for governance purposes.

Fourth, we considered **project scope and MVP delivery**. Since the first goal is to produce a workable recruitment system, features that directly support the end-to-end hiring process were prioritised above reporting and auditing functions. As a result, audit logs and report export were placed in the low or very low categories.

This prioritisation method produced a backlog that is both practical and logically staged. It ensures that the team focuses first on the features most critical to usability and project success.

1.4 Estimation method

For effort estimation, the team used **story points** rather than time-based estimates. Story points were chosen because they are more appropriate for early-stage planning, when exact development time is still uncertain. Instead of predicting the number of hours required, story points allow the team to estimate requirements comparatively based on relative effort.

The estimates in the backlog reflect several factors: implementation complexity, uncertainty, dependency on other features, and the amount of validation required. Simple stories such as login or viewing basic summary information were given relatively small estimates, while more complex stories such as applying for a job, accepting or rejecting applicants, or generating workload analysis received higher estimates.

The team treated estimation as a **relative judgement process**. Stories were compared against one another rather than estimated in isolation. For example, a login feature was considered much simpler than an application workflow, while application status tracking was considered

less complex than applicant selection and workload detection. This comparative approach helped maintain internal consistency across the backlog.

Story point estimation also supports sprint planning more effectively than time estimation at this stage. Since the team can measure sprint capacity in terms of relative workload, it becomes easier to distribute stories across iterations without assuming unrealistic precision. In this project, the estimates helped us balance the sprints so that foundational work appears early, core workflow stories are grouped in the middle, and administrative enhancements are scheduled later.

In summary, story points were used because they provide a practical and flexible estimation method for an initial backlog. They support prioritisation, iteration planning, and scope control while remaining suitable for a student project environment where requirements may still evolve.

2. Supporting material

Include supporting materials, such as interview questions, survey results, and customer feedback etc (if applicable).

2. Supporting material

2.1 Interview Questions

Questionnaires

Q1. Primary Role Survey

- Purpose: Classify user types to support demand stratification.
- Options: Teaching Assistant (TA), Module Organiser (MO), Administrator (Admin), Other.

TA Recruitment System
Requirements Survey

*1.What is your primary role?

☐ A.Teaching Assistant (TA)
☐ B.Module Organiser (MO)
☐ C.Administrator (Admin)
☐ D.Other:

Q2. System Usage Frequency Survey

- Purpose: Assess system usage intensity to guide performance and iteration planning.
- Options: Multiple times per week, Once a week, Several times per month, Rarely.

•2.How often do you (or would you) use the recruitment system?

☐ A.Multiple times per week
☐ B.Once a week
☐ C.Several times per month
☐ D.Rarely

Q3. TA Feature Importance Rating (1–5 scale)

- Core Needs: Account registration/login, applicant profile management, job application operations (CV upload, application submission, status tracking), application outcome notifications, job matching recommendations, and application history with past evaluations.

•3.(TA Features)Please rate how important each of the following features is to you on a scale of 1 to 5 (1 = Not needed at all, 5 = Absolutely essential).

1	2	3	4	5
Register account (using university email)				
☐	☐	☐	☐	☐
Log in to the system				
☐	☐	☐	☐	☐
Create applicant profile (skills, availability, etc.)				
☐	☐	☐	☐	☐
Edit applicant profile				
☐	☐	☐	☐	☐

1	2	3	4	5
Upload CV				
☐	☐	☐	☐	☐
Search and filter jobs				
☐	☐	☐	☐	☐
Apply for a job				
☐	☐	☐	☐	☐
Track application status				
☐	☐	☐	☐	☐

1	2	3	4	5
Receive notifications of application outcome				
☐	☐	☐	☐	☐
Job matching recommendations				
☐	☐	☐	☐	☐
Application history and past evaluations				
☐	☐	☐	☐	☐

Q4. MO Feature Importance Rating (1–5 scale)

- Core Needs: Job posting creation/management, applicant shortlisting/approval, batch operations (filtering, interview invitation sending), and TA performance evaluation records.

•4.(MO Features)Please rate how important each of the following features is to you on a scale of 1 to 5 (1 = Not needed at all, 5 = Absolutely essential).

1	2	3	4	5
Log in as Module Organiser				
☐	☐	☐	☐	☐
Create a job posting				
☐	☐	☐	☐	☐
Edit a job posting				
☐	☐	☐	☐	☐
Publish / close a job posting				
☐	☐	☐	☐	☐

1	2	3	4	5
View applicants for a job				
☐	☐	☐	☐	☐
Shortlist applicants				
☐	☐	☐	☐	☐
Accept or reject applicants				
☐	☐	☐	☐	☐
Batch filtering, batch sending of interview invitations				
☐	☐	☐	☐	☐
TA performance evaluation and records				
☐	☐	☐	☐	☐

Q5. Admin Feature Importance Rating (1–5 scale)

- Core Needs: TA workload monitoring, application statistics, user account management, system notifications and permission configuration, and CSV workload report export.

* 5.(Admin Features)Please rate how important each of the following features is to you on a scale of 1 to 5 (1 = Not needed at all, 5 = Absolutely essential).

	1	2	3	4	5
Log In as Admin	☐	☐	☐	☐	☐
View all TA workloads	☐	☐	☐	☐	☐
Filter workload by term or module	☐	☐	☐	☐	☐
Detect overloaded TAs	☐	☐	☐	☐	☐

	1	2	3	4	5
View application statistics (total, accept/reject)	☐	☐	☐	☐	☐
Manage user accounts (activate/suspend)	☐	☐	☐	☐	☐
Reset admin password (secure recovery)	☐	☐	☐	☐	☐
Export workload report (CSV)	☐	☐	☐	☐	☐
Publish system notifications or announcements to specific roles	☐	☐	☐	☐	☐
Support custom permissions for different administrator roles (e.g., read-only, editable, exportable) to facilitate tiered management.	☐	☐	☐	☐	☐

Q6. Interface Preference Survey

- Options: Clean and task-oriented, rich in data visualization, customizable layout, no preference (as long as functional).

* 6.What kind of interface do you prefer?

☐ Clean and functional, task-oriented

☐ Rich in charts and data visualization

☐ Customizable layout

☐ No preference, as long as it works well

Q7. Current Process Pain Points Survey

- Key Issues: Lack of built-in communication channels, absence of intelligent job matching, no historical performance records, heavy manual repetitive work, and difficulty tracking application status across modules.

* 7.What are the top three problems you have encountered (or anticipated) with current recruitment processes/systems? [Multiple]

☐ No built-in communication between MOn and TAs; everything relies on external email

☐ No automatic job matching based on skills, availability, or past experience

☐ Lack of past performance records for TA applicants

☐ Manual and repetitive work when handling large numbers of applications (e.g., sending invitations one by one)

☐ Difficulty tracking application status across multiple modules

Q8. Desired New Features Survey

- Top Requests: Job matching based on TA profiles, application history with evaluations, batch operations, and interview schedule integration.

* 8.What are the top three features you would most like to see added? [Multiple]

☐ Job matching recommendations based on TA profile (skills, availability, experience)

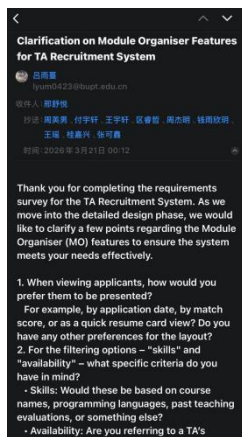
☐ Application history with past feedback/evaluations for TAs

☐ Batch operations (e.g., bulk filtering, bulk invitations, bulk accept/reject)

☐ Interview scheduling and confirmation within the system

Submit

Email Enquiries



2.2 Survey Results

TA Recruitment System Requirements Survey

1.What is your primary role? [Single Choice]

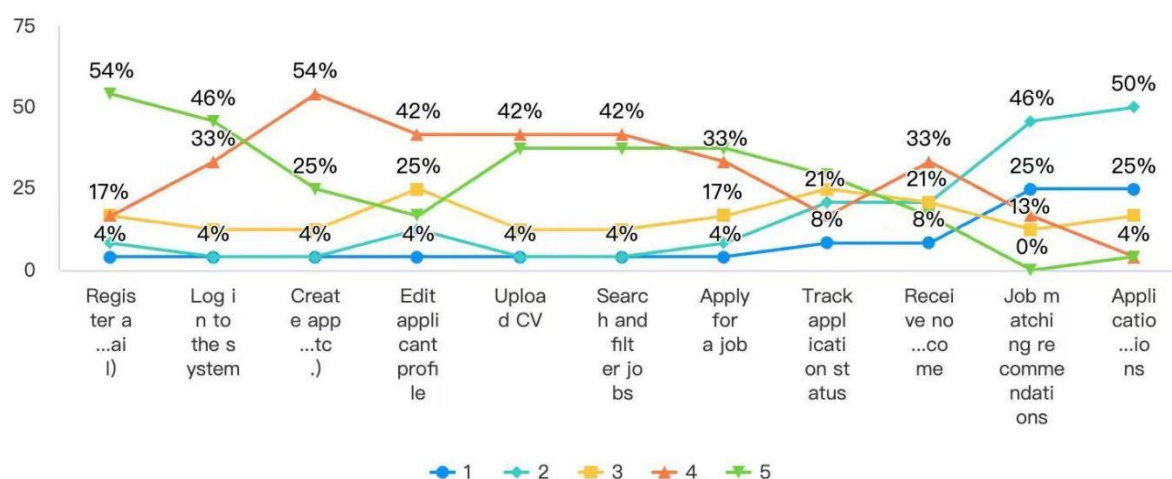
Option	Subtotal	Percentage
A.Teaching Assistant (TA)	12	50%
B.Module Organiser (MO)	8	33.33%
C.Administrator (Admin)	4	16.67%
D.Other:	0	0%
Valid Count Per Participant	24	

2.How often do you (or would you) use the recruitment system? [Single Choice]

Option	Subtotal	Percentage
A.Multiple times per week	8	33.33%
B.Once a week	9	37.5%
C.Several times per month	7	29.17%
D.Rarely	0	0%
Valid Count Per Participant	24	

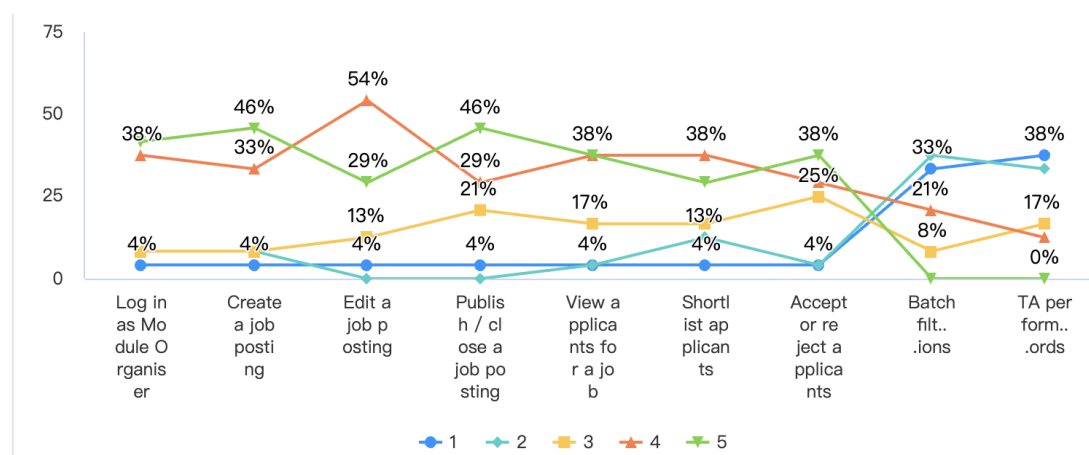
3.(TA Features)Please rate how important each of the following features is to you on a scale of 1 to 5 (1 = Not needed at all, 5 = Absolutely essential). [Matrix Radio Button]

Question\Option	1	2	3	4	5
Register account (using university email)	1(4.17%)	2(8.33%)	4(16.67%)	4(16.67%)	13(54.17%)
Log in to the system	1(4.17%)	1(4.17%)	3(12.5%)	8(33.33%)	11(45.83%)
Create applicant profile (skills, availability, etc.)	1(4.17%)	1(4.17%)	3(12.5%)	13(54.17%)	6(25%)
Edit applicant profile	1(4.17%)	3(12.5%)	6(25%)	10(41.67%)	4(16.67%)
Upload CV	1(4.17%)	1(4.17%)	3(12.5%)	10(41.67%)	9(37.5%)
Search and filter jobs	1(4.17%)	1(4.17%)	3(12.5%)	10(41.67%)	9(37.5%)
Apply for a job	1(4.17%)	2(8.33%)	4(16.67%)	8(33.33%)	9(37.5%)
Track application status	2(8.33%)	5(20.83%)	6(25%)	4(16.67%)	7(29.17%)
Receive notifications of application outcome	2(8.33%)	5(20.83%)	5(20.83%)	8(33.33%)	4(16.67%)
Job matching recommendations	6(25%)	11(45.83%)	3(12.5%)	4(16.67%)	0(0%)
Application history and past evaluations	6(25%)	12(50%)	4(16.67%)	1(4.17%)	1(4.17%)



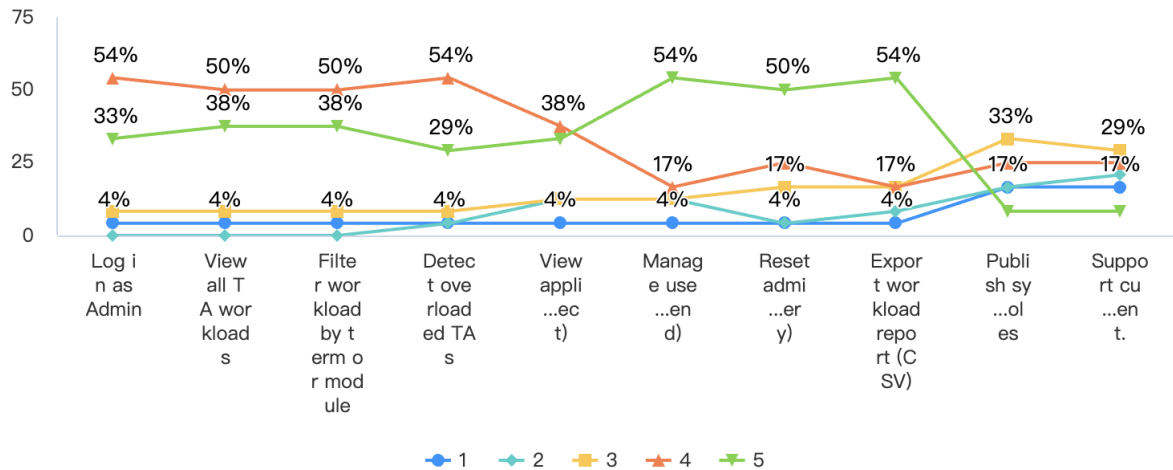
4.(MO Features)Please rate how important each of the following features is to you on a scale of 1 to 5 (1 = Not needed at all, 5 = Absolutely essential). [Matrix Radio Button]

Question\Option	1	2	3	4	5
Log in as Module Organiser	1(4.17%)	2(8.33%)	2(8.33%)	9(37.5%)	10(41.67%)
Create a job posting	1(4.17%)	2(8.33%)	2(8.33%)	8(33.33%)	11(45.83%)
Edit a job posting	1(4.17%)	0(0%)	3(12.5%)	13(54.17%)	7(29.17%)
Publish / close a job posting	1(4.17%)	0(0%)	5(20.83%)	7(29.17%)	11(45.83%)
View applicants for a job	1(4.17%)	1(4.17%)	4(16.67%)	9(37.5%)	9(37.5%)
Shortlist applicants	1(4.17%)	3(12.5%)	4(16.67%)	9(37.5%)	7(29.17%)
Accept or reject applicants	1(4.17%)	1(4.17%)	6(25%)	7(29.17%)	9(37.5%)
Batch filtering, batch sending of interview invitations	8(33.33%)	9(37.5%)	2(8.33%)	5(20.83%)	0(0%)
TA performance evaluation and records	9(37.5%)	8(33.33%)	4(16.67%)	3(12.5%)	0(0%)



5.(Admin Features)Please rate how important each of the following features is to you on a scale of 1 to 5 (1 = Not needed at all, 5 = Absolutely essential). [Matrix Radio Button]

Question\Option	1	2	3	4	5
Log in as Admin	1(4.17%)	0(0%)	2(8.33%)	13(54.17%)	8(33.33%)
View all TA workloads	1(4.17%)	0(0%)	2(8.33%)	12(50%)	9(37.5%)
Filter workload by term or module	1(4.17%)	0(0%)	2(8.33%)	12(50%)	9(37.5%)
Detect overloaded TAs	1(4.17%)	1(4.17%)	2(8.33%)	13(54.17%)	7(29.17%)
View application statistics (total, accept/reject)	1(4.17%)	3(12.5%)	3(12.5%)	9(37.5%)	8(33.33%)
Manage user accounts (activate/suspend)	1(4.17%)	3(12.5%)	3(12.5%)	4(16.67%)	13(54.17%)
Reset admin password (secure recovery)	1(4.17%)	1(4.17%)	4(16.67%)	6(25%)	12(50%)
Export workload report (CSV)	1(4.17%)	2(8.33%)	4(16.67%)	4(16.67%)	13(54.17%)
Publish system notifications or announcements to specific roles	4(16.67%)	4(16.67%)	8(33.33%)	6(25%)	2(8.33%)
Support custom permissions for different administrator roles (e.g., read-only, editable, exportable) to facilitate tiered management.	4(16.67%)	5(20.83%)	7(29.17%)	6(25%)	2(8.33%)



6. What kind of interface do you prefer? [Single Choice]

Option	Subtotal	Percentage
Clean and functional, task-oriented	8	33.33%
Rich in charts and data visualization	11	45.83%
Customizable layout	3	12.5%
No preference, as long as it works well	2	8.33%
Valid Count Per Participant	24	

7. What are the top three problems you have encountered (or anticipate) with current recruitment processes/systems? [Multiple Choice]

Option	Subtotal	Percentage
No built-in communication between MOs and TAs; everything relies on external email	20	83.33%
No automatic job matching based on skills, availability, or past experience	20	83.33%
Lack of past performance records for TA applicants	15	62.5%
Manual and repetitive work when handling large numbers of applications (e.g., sending invitations one by one)	14	58.33%
Difficulty tracking application status across multiple modules	3	12.5%
Valid Count Per Participant	24	

8. What are the top three features you would most like to see added? [Multiple Choice]

Option	Subtotal	Percentage
Job matching recommendations based on TA profile (skills, availability, experience)	18	75%
Application history with past feedback/evaluations for TAs	20	83.33%
Batch operations (e.g., bulk filtering, bulk invitations, bulk accept/reject)	21	87.5%
Interview scheduling and confirmation within the system	13	54.17%
Valid Count Per Participant	24	

2.3 UML diagram version1

